STANDATA interpretation 23-BCI-032/23-FCI-005

Building/Fire

Occupancy classification for buildings used for tank vehicles

Date Issued: July 2025

Purpose

To clarify the application of the National Building Code – 2023 Alberta Edition (NBC(AE)) and the National Fire Code – 2023 Alberta Edition (NFC(AE)) to buildings used for parking, repairing, and/or servicing of tank vehicles.

Discussion

Tank vehicles containing flammable liquids or vapours, or combustible liquids or vapours present a unique hazard inside a building. In addition to the hazard inherent in the product itself, there is a potential for the release of flammable or combustible vapours in significant quantities which can overwhelm a standard ventilation system for a building and become a fire or explosion risk.

The activity of bringing a tank vehicle into a building involves specific references from both the NBC(AE) and the NFC(AE). The Provincial Administrators for Building and Fire consider a joint interpretation necessary to communicate the importance of applying this interpretation to both new building construction and practices within existing buildings. The NBC(AE) applies to:

1. new buildings prior to occupancy; or
2. existing buildings undergoing a change in occupancy.

The NFC(AE) applies to the operation and maintenance of existing buildings and establishes requirements for tank vehicles.

Code References

NFC(AE) Division A, Article 1.4.1.2.

1.4.1.2. Defined Terms

Authority having jurisdiction means a safety codes officer in the fire discipline exercising authority pursuant to designation of powers and terms of employment in accordance with the Safety Codes Act.

Building means any structure used or intended for supporting or sheltering any use or *occupancy*.

Combustible liquid means a liquid having a *flash point* at or above 37.8°C and below 93.3°C. (See Subsection 4.1.2. of Division B.)

Flammable liquid means a liquid having a *flash point* below 37.8°C and having a vapour pressure not more than 275.8 kPa (absolute) at 37.8°C as determined by ASTM D323,"Standard Test Method for Vapor Pressure of Petroleum Products (Reid Method)." (See Subsection 4.1.2. of Division B.)

Tank vehicle means any vehicle, other than railroad tank cars and boats, with a cargo tank having a capacity of more than 450 L, mounted or built as an integral part of the vehicle and used for the transportation of *flammable liquids* or *combustible liquids* and including tank trucks, trailers and semi-trailers.

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NFC(AE) Division B, Article 4.11.2.3.

4.11.2.3. Parking inside Buildings

1) A *tank vehicle* shall not be parked inside a *building* unless

a) the *building* is specifically designed for that purpose,

b) there is sufficient space in the *tank vehicle* compartment to accommodate the thermal expansion of the *flammable liquid* or *combustible liquid*, and

c) the *tank vehicle* has no leaks of *flammable liquid* or *combustible liquid*.

NFC(AE) Division B, Article 4.11.2.5.

4.11.2.5. Repair and Servicing

1) No person shall repair or service the cargo tank of a *tank vehicle* in a *building* unless

a) all *flammable liquids* and *combustible liquids* have been removed from the cargo tank and stored in a *storage tank* or container constructed and installed in accordance with this Part,

b) the cargo tank has been purged with steam of flammable vapours or combustible vapours, outside the *building* or in a well-ventilated area, to less than 10% of the lower explosive limit, and

c) while the cargo tank is being repaired, it is monitored for the presence of flammable vapours and combustible vapours using combustible gas detection equipment.

2) No person shall repair, service or store a *tank vehicle* in a *building* other than for repairs conducted in conformance with Article 4.11.2.2. and Sentence (1), unless

a) the *building* is acceptable to the *authority having jurisdiction*, or

b) all *flammable liquids* and *combustible liquids* have been removed from the cargo tank and the tank has been purged of flammable vapours or combustible vapours, outside the *building* or in a well-ventilated area, to less than 10% of the *lower explosive limit*.

NBC(AE) Division B, Sentence 3.2.2.6.(1)

3.2.2.6. Multiple Major Occupancies

1) Except as permitted by Articles 3.2.2.7. and 3.2.2.8., and Sentences 3.2.2.48.(4), 3.2.2.51.(5), 3.2.2.57.(3) and 3.2.2.60.(4), in a *building* containing more than one *major occupancy*, the requirements of this Subsection for the most restricted *major occupancy* contained shall apply to the whole *building*.

NBC(AE) Division B, Sentence 3.2.2.8.(1)

3.2.2.8. Exceptions for Major Occupancies

1) In a *building* in which the aggregate area of all *major occupancies* in a particular Group or Division is not more than 10% of the *floor area* of the *storey* in which they are located, these *major occupancies* need not be considered as *major occupancies* for the purposes of this Subsection, provided they are not classified as Group F, Division 1 or 2 *occupancies*.

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Application

This interpretation applies to buildings used for any one or all of the following activities: parking, repairing, and servicing of tank vehicles.

Interpretation

1. Buildings used for parking tank vehicles containing flammable or combustible gases or vapours are classified as Group F, Division 1, high hazard industrial occupancies and must:

- be specifically designed for that purpose according to both the NBC(AE) and the NFC(AE);
- be provided with a ventilation system in accordance with Part 6 of the NBC(AE);
- have electrical systems installed in conformance with the electrical code regulation made pursuant to the Safety Codes Act;
- have fuel-fired appliances in service rooms with no openings between the hazardous location and the components capable of igniting a flammable or combustible vapour; and
- have fuel-fired appliances installed in conformance with the applicable codes and standards, and in the case of gas-fired appliances, in conformance to the gas code regulation made pursuant to the Safety Codes Act.

Note: Additional requirements of Part 3 or Part 6 of the NBC(AE) or Part 4 of the NFC(AE) may apply.

2. Buildings to be used for parking, repairing, and servicing tank vehicles that have been purged of their flammable or combustible gases or vapours are classified as Group F, Division 2, medium hazard industrial occupancies, provided:

- the area is specifically designed for that purpose;
- all flammable liquids and combustible liquids have been removed and the tanks are purged in accordance with Section 4.11. of the NFC(AE);
- there is a ventilation system in accordance with Part 6 of NBC(AE);
- electrical systems are installed in conformance with the electrical code regulation made pursuant to the Safety Codes Act;
- fuel-fired appliances are installed in service rooms and have no openings between the garage location and the components capable of igniting a flammable or combustible vapour unless the appliances are at a height as permitted by the applicable codes or regulation; and
- fuel-fired appliances installed in conformance with the applicable codes, and in the case of gas-fired appliances in conformance to the gas code regulation made pursuant to the Safety Codes Act.

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Fire Code – 2023 Alberta Edition

Issued by the Provincial Fire and Building Administrators

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